Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

Wireshark Practical: Capturing Network Packets

Aim: To capture network packets using Wireshark and identify the protocols used when visiting a website.

I can't run Wireshark on your laptop directly, but follow these steps to complete the practical.

1. Steps to perform

1. Download Wireshark from the official website: Wireshark Download.
2. Install it on your laptop.
3. Open Wireshark and select your active Wi-Fi or Ethernet network interface.
4. Click Start Capturing Packets.
5. Open your browser and visit `https://example.com`.
6. Return to Wireshark and click the red Stop button.
7. Look at the Protocol column in the packet list.

2. Sample observation table

Your actual results may differ depending on the website and network.

No.	Protocol	Purpose (Easy Language)
1	DNS	Finds the IP address of a website using its name.
2	TCP	Provides reliable data delivery between devices.
3	TLS	Helps encrypt and protect HTTPS communication.
4	HTTP	Carries web requests and responses; HTTPS traffic is encrypted.
5	QUIC	Supports fast, secure web communication, often over UDP.
6	UDP	Sends datagrams used by protocols such as DNS and QUIC.
7	ARP	Helps find a device's MAC address on a local IPv4 network.

Note: You might see `TLS`, `QUIC`, `TCP`, or `UDP` instead of `HTTP`, because secure website traffic is encrypted and modern browsers may use QUIC.

3. How to filter packets

In Wireshark's display filter bar, try:

- `dns` — shows DNS packets.

- `tcp` — shows TCP packets.
- `tls` — shows TLS packets.
- `quic` — shows QUIC packets, if present.
- `http` — shows packets Wireshark identifies as HTTP.

Press Enter after typing a filter.

4. Result

After capturing packets while visiting a website, I can identify protocols such as DNS, TCP, TLS, UDP, and possibly QUIC in Wireshark's packet list.

Conclusion: Wireshark helps us understand how a computer communicates with websites by capturing and displaying network packets and their protocols.

**Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.).

Hint: Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.**

Wireshark Practical: Mapping Packets to OSI Layers

Aim: To identify the OSI layers represented by three packets captured in Wireshark.

Here is a sample observation table. Your actual packets may be different.

Packet	Protocol	OSI Layer	Reason
1	DNS	Layer 7 – Application	DNS finds the IP address of a website using its name.
2	TCP	Layer 4 – Transport	TCP helps deliver data reliably between devices.
3	IP	Layer 3 – Network	IP addresses help route packets to the correct destination.

How to check in Wireshark

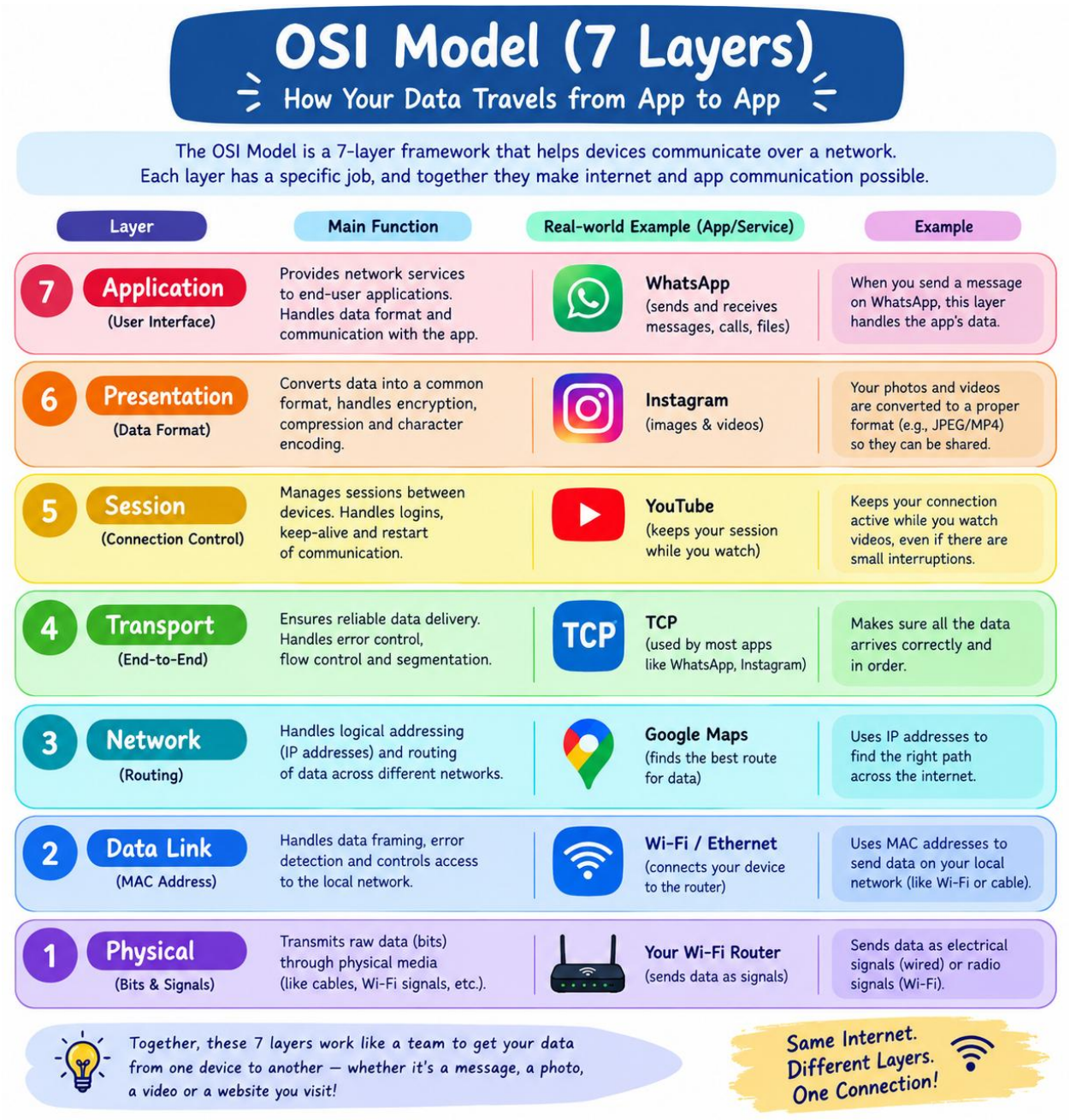
1. Open your captured file in Wireshark.
2. Look at the Protocol column.
3. Click any packet to select it.
4. Expand the packet details to see protocols such as DNS, TCP, or IP.
5. Match each protocol to its main OSI layer using the table above.

Conclusion

Wireshark helps us identify protocols used for network communication. DNS works at the Application layer, TCP at the Transport layer, and IP at the Network layer.

Note: These are sample packets. Use three packets from your own capture for your final practical record.

Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).



**Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

Hint: Think about how your message is typed, packaged, sent over the internet, and received.**

How a WhatsApp Message Travels Through the OSI Model

When I send a message like “Hello” on WhatsApp, it passes through different OSI layers to reach my friend's phone.

1. Application Layer (Layer 7)

I type “Hello” and press Send in WhatsApp. The app prepares the message for communication.

2. Presentation Layer (Layer 6)

The data is represented in a suitable format and protected using encryption. WhatsApp uses end-to-end encryption for personal messages.

3. Transport Layer (Layer 4)

Transport protocols help carry the data between your device and the service. TCP or other transport mechanisms may be used, depending on the connection.

4. Network Layer (Layer 3)

IP addresses help routers forward the data across networks toward the destination service.

5. Data Link and Physical Layers (Layers 2 and 1)

Wi-Fi or Ethernet carries the data across the local network. Radio signals or electrical signals transmit the bits.

What happens at the receiver's phone?

The data travels through the network to WhatsApp's service and then to your friend's phone. The receiving device processes the data through the network layers, decrypts the message on the recipient's device, and displays “Hello” in the WhatsApp chat.

Conclusion: The OSI model helps us understand how a WhatsApp message is prepared, protected, transmitted across networks, and delivered to the receiver. In real networks, the OSI layers are a learning model, and protocols do not always map perfectly to one layer.